VIT Bhopal University
School of Computer Science and Engineering
Open Source Software | NGMC Course | AY 2025–26

The Open Source Audit

A Capstone Project Report

Chosen Software: **Git** — Distributed Version Control System (GPL v2)

The Open Source Audit: Git

Course: Open Source Software (NGMC) | VIT Bhopal University

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1. Abstract

This report provides a structured audit of Git, the world's most widely used version control system. It covers the origin story behind Git's creation, the legal implications of the GPL v2 license, the ethics of open infrastructure, and a technical analysis of Git's footprint on Linux systems. Additionally, the report includes five custom Bash scripts designed to demonstrate practical Linux administration and automation.

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2. Origin and Philosophy

2.1 The Crisis of 2005

Before Git, the Linux kernel team relied on a proprietary tool called BitKeeper. In 2005, a licensing dispute led the vendor to revoke access for the entire team overnight. Linus Torvalds decided that the kernel should never again depend on a tool that could be taken away.

2.2 The Ten-Day "Rage-Code"

In roughly ten days, Torvalds wrote the initial version of Git. He designed it to be the exact opposite of existing centralized systems like CVS. The core goals were:

- **Speed: Handling massive amounts of data instantly.**
- **Distributed Architecture: Every developer owns a full copy of the history.**
- **Integrity: Using cryptographic hashing to ensure history cannot be secretly altered.**

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3. The Legal Shield: GPL v2

3.1 The Four Freedoms

Git is governed by the GNU General Public License v2. This license ensures four fundamental rights:

1. **Freedom 0: Run the software for any purpose.**
2. **Freedom 1: Study and modify the code (requires source access).**
3. **Freedom 2: Redistribute copies freely.**
4. **Freedom 3: Share modified versions.**

3.2 Copyleft vs. Permissive Licenses

Unlike MIT licenses, which allow code to be "closed" in derivative works, GPL v2 includes a copyleft clause. If you distribute a modified version of Git, you must release those changes under the same open license. This prevents companies from monetizing community improvements while keeping them secret.

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4. Technical Footprint on Linux

4.1 Filesystem Hierarchy

When Git is installed (via apt or dnf), it maps to specific system directories:

- **/usr/bin/git: The main executable.**
- **/usr/lib/git-core/: Contains sub-commands like git-fetch and git-merge.**
- **~/.gitconfig: Per-user configuration for names and emails.**
- **.git/: The hidden directory in any repository containing the full object store and history.**

4.2 Security Architecture

Git follows the Principle of Least Privilege. It does not run as a permanent system daemon; instead, it runs with the permissions of the user who invokes it. This ensures that a compromise in a Git process cannot easily gain root access to the machine.

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5. Comparative Analysis: Git vs. Perforce

For this audit, Git was compared against Perforce Helix Core, its primary proprietary rival.

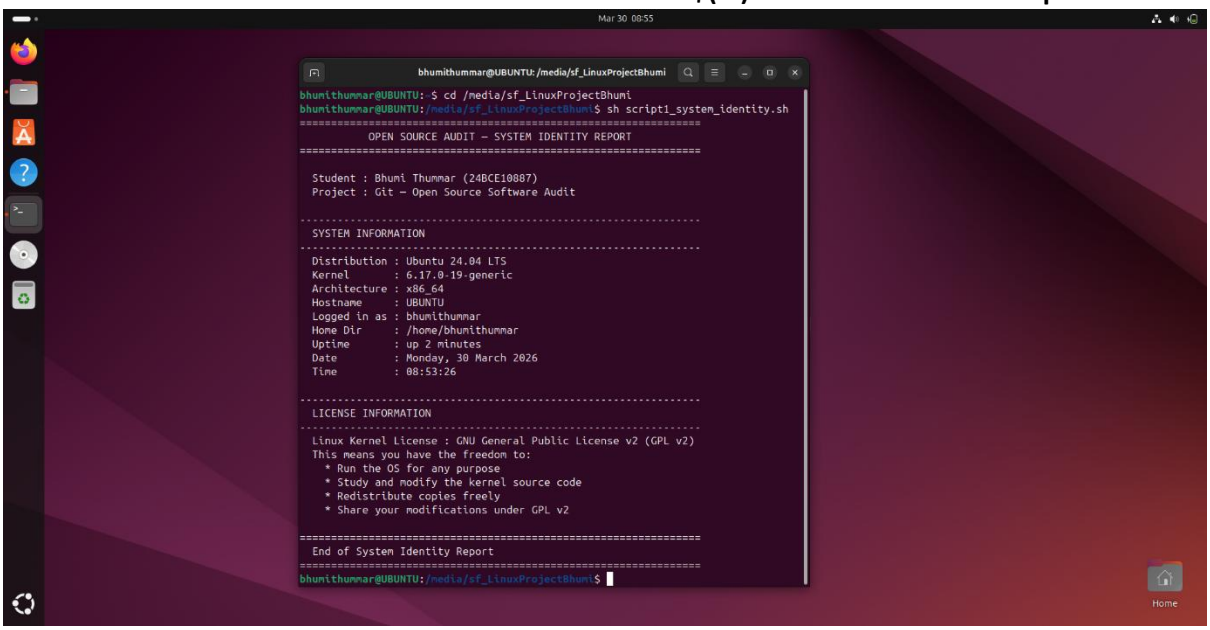
Dimension	Git (GPL v2)	Perforce Helix Core
Cost	Free and community-owned.	Tens of thousands in fees.
Transparency	Full source is public and auditable.	Closed source; trust the vendor.
Governance	Software Freedom Conservancy.	Corporate-driven roadmap.
Strength	Branching, merging, speed.	Large 4K/Binary file assets.

6. Practical Implementation: Shell Scripts

Five scripts were developed to demonstrate Linux automation skills.

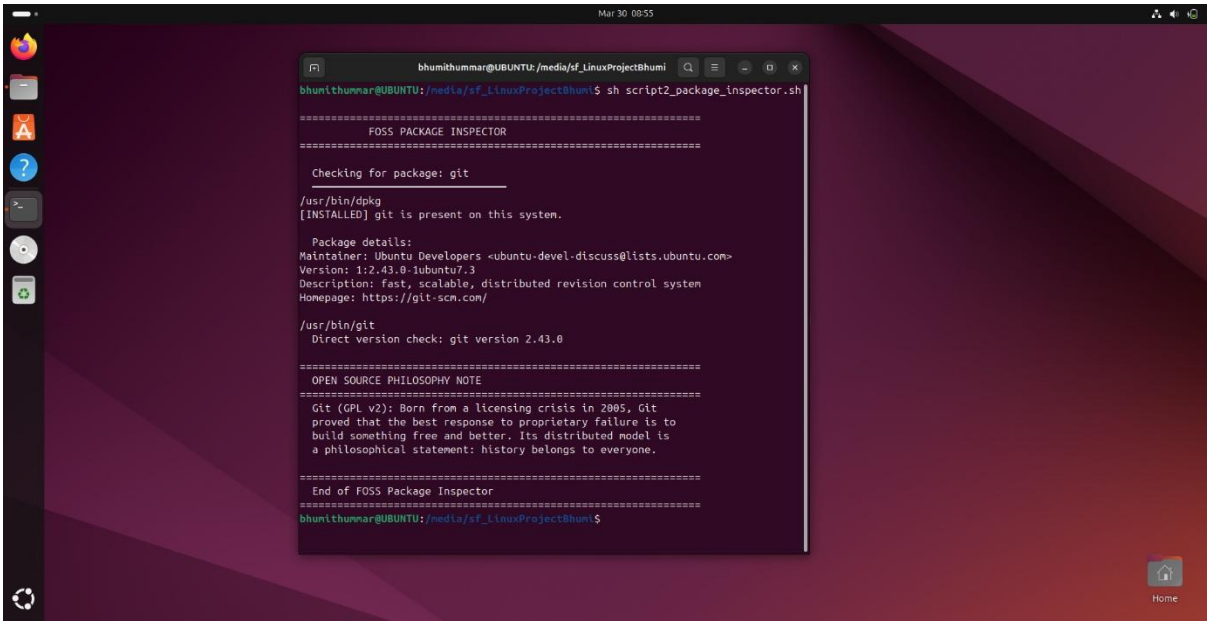
6.1 Script 1: System Identity Report

- **Function: Captures kernel version (uname -r), architecture, and uptime.**
- **Educational Value: Demonstrates command substitution $(...) and formatted shell output.**



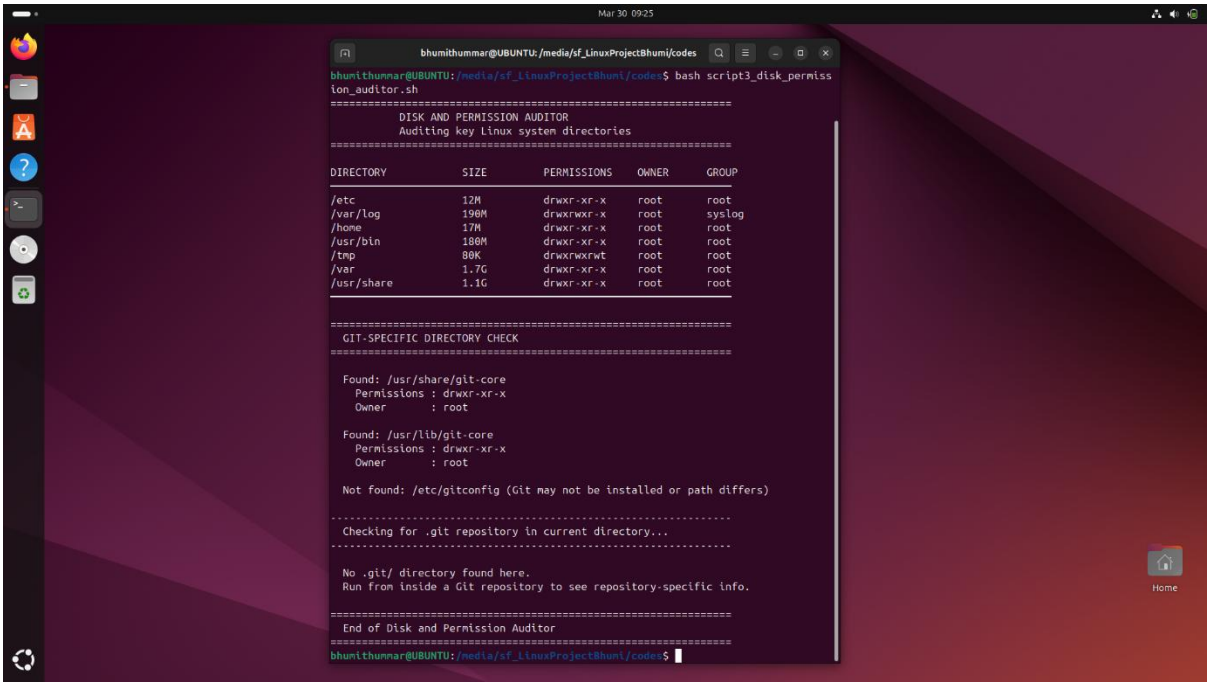
6.2 Script 2: FOSS Package Inspector

- **Function: Detects if the system uses dpkg or rpm and checks for Git installation.**
- **Educational Value: Showcases nested if-else logic and case statements.**



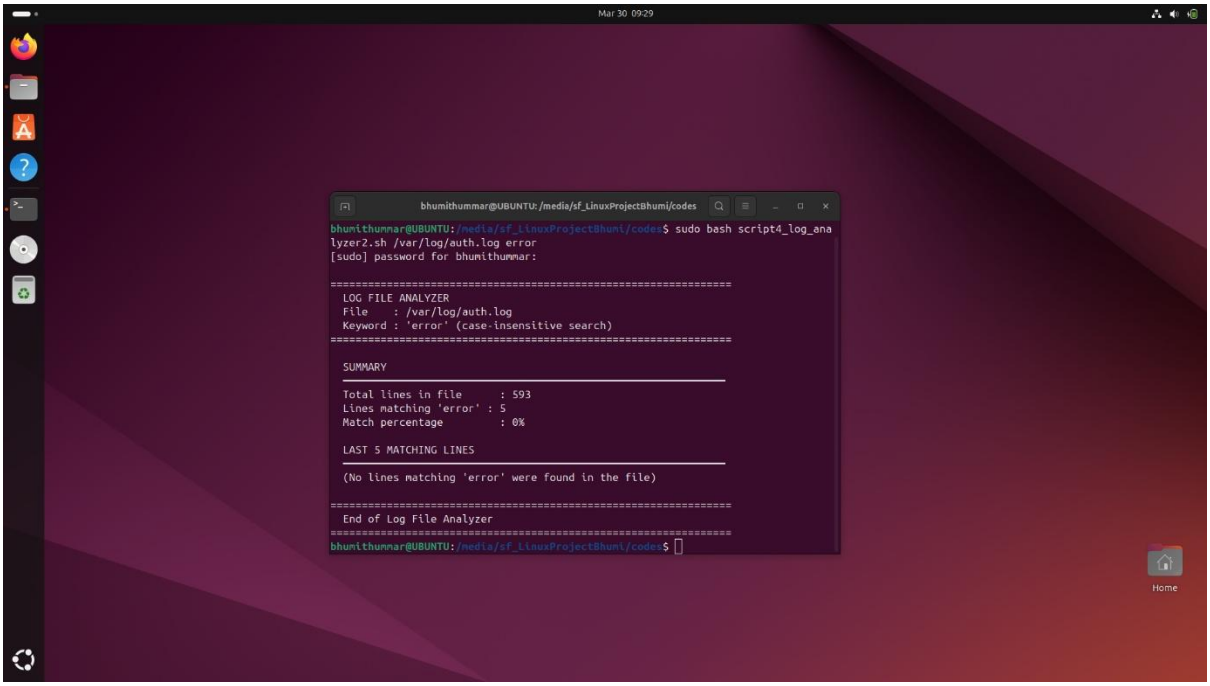
6.3 Script 3: Disk & Permission Auditor

- **Function: Iterates through an array of directories to report size and permissions.**
- **Educational Value: Uses awk to extract specific metadata fields from ls -ld.**



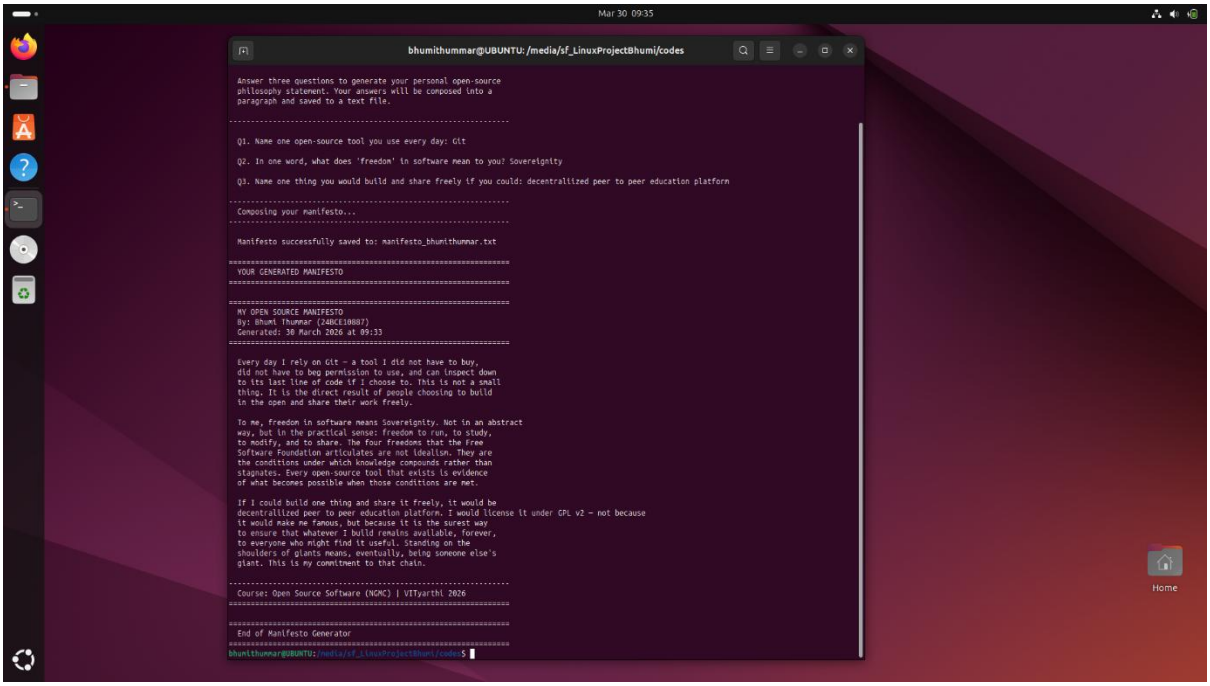
6.4 Script 4: Log File Analyzer

- **Function: Reads log files line-by-line to find keywords (like "error") and calculates a match percentage.**
- **Educational Value: Demonstrates integer arithmetic and while read loops.**



6.5 Script 5: Manifesto Generator

- **Function: An interactive script that captures user values and saves them to a .txt file.**
- **Educational Value: Covers user input validation and file redirection (>>).**



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7. Conclusion

Git is the connective tissue of modern software delivery. Its success was not accidental; it was a direct result of the design philosophy and the protection of the GPL v2 license. While proprietary tools have niches, Git remains the superior choice for most projects because it is "locked open"—impossible to take away from the community that relies on it.

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8. References

1. **GNU Project: The Free Software Definition.**
2. **Open Source Initiative (OSI): Licensing Standards.**
3. **Eric S. Raymond: *The Cathedral and the Bazaar*.**
4. **Git Reference Manual: Documentation and workflows.**
5. **William Shotts: *The Linux Command Line*.**